

EXPLORE / DISCOVERY probe freely · inspect many signals · form hypotheses**M1****Suspicious Behavior Detection**

Run the analyzer probing library, find per-case suspicious behaviors

BLACK-BOX BEHAVIOR

SELECTED

- consistency across 5 resamples
- coverage / verification gap
- answer-extraction audit

INTERNAL / WHITE-BOX

- attention & attribution
- logits & representations

MULTIMODAL

- grounding & hallucination

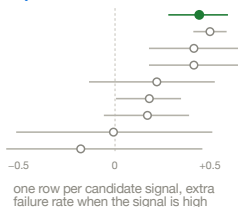
What the Probes Ask?

- Did it answer in the form we asked for?
- Did it give up answering?
- Same question five times, same answer?
- Reworded question, same answer?
- Did it ever produce the right answer?

26 measurements, 9 forwarded to M2

M2**Statistical Screening**

Using plots to explain, statistical tests to decide

Forest plot**Statistical Tests**

1 of 9 signals survives Benjamini-Hochberg (BH)

+0.44, CI [+0.28, +0.59], $p = 9.7e-7$

M3**Hypothesis Formation**

Explore the reason behind the signals

THE THREE HYPOTHESES**H1 Calculation error**

Predictions give difference, ratio or sum

H2 Consistent error over datasets

5 identical answers, all are wrong

H3 Language prior

The model overlooks the image, directly answers from its parametric knowledge

M5**Held-out Verification**

Confirm whether the hypothesis is verified over the held-out cases

SUPPORTED

H1 calculation error

SUPPORTED

H2 Consistent error over datasets

INCONCLUSIVE

H3 language prior

M4**Validated Repair**

Fix the failure with the validated repair ladder

L1

Prompt / instructions

Non-significant +0.01

L2

Scaffold / tools / multi-call

Accepted +0.28

L3

Internals / read & write

Regressed -0.06

L4

Re-training

Untouched

6 CANDIDATES TRIED ON EXPLORE

L1 visual_grounding

L2 grounded_upscale

L2 rebind_verify

L2 transcribe_zoom

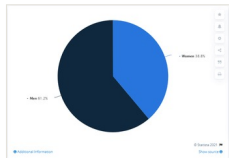
L2 self_consistency

L3a coded_pipeline

-0.05 0 +0.1 +0.2

The accepted repair — transcribe_then_refine_zoom (L2 scaffold)

Example: chartqa-human-1142 — Q: Work out the ratio of the bigger segment to the smaller one? A: 1.577

**1. Image operation**

crop salient region
upscale x2
boost contrast x1.4

Better visual grounding
before asking anything

2. Forced read-out

Men -> 61.2%
Women -> 38.8%

Transcribe every data mark as label -> value

3. Self-check

- Are two rows swapped?
- Is a value on the wrong label?
- Is an end-of-axis mark missing?

4. Answer

from the corrected list only

1.577

3 samples, modal answer

Over the full held-out testing cases

Baseline

49.2%

with the repair

79.7%

45 fixed, 6 broken, gain far beyond chance ($p < .001$)